TOURISM PACKAGE AND TRAVEL BOOKING MANAGEMENT SYSTEM

DBMS Mini Project — Review 1

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PROBLEM IDENTIFICATION

Spreadsheets cause data duplication and errors.

Manual booking leads to capacity conflicts and overbooking.

Tracking guides, cancellations, and refunds is inefficient.

Report generation on revenue and destination demand is difficult.

THE NEED

A centralized, efficient, accurate, and secure database system for tourism packages and bookings.



Project Objectives

- Design a relational database for tourism and travel management.
- Store and manage package, destination and itinerary details.
- Maintain departure schedules, hotel details, transport and guide assignments.
- Manage customers, travellers, bookings and payment records.
- Handle cancellations and refunds accurately.
- Prevent overbooking and invalid data through database constraints.
- Generate useful reports using SQL queries.
- Develop a simple front-end application using Java or Python.

Project Scope

Package Management

Create tour packages with destinations, itinerary days, hotels and transport information.

Booking Management

Register customers and travellers, make bookings and check available seats.

Departure and Guide Management

Schedule departures, assign guides and avoid overlapping assignments.

Payments and Reports

Record payments, cancellations and refunds, and generate business reports.

Users and Functional Requirements

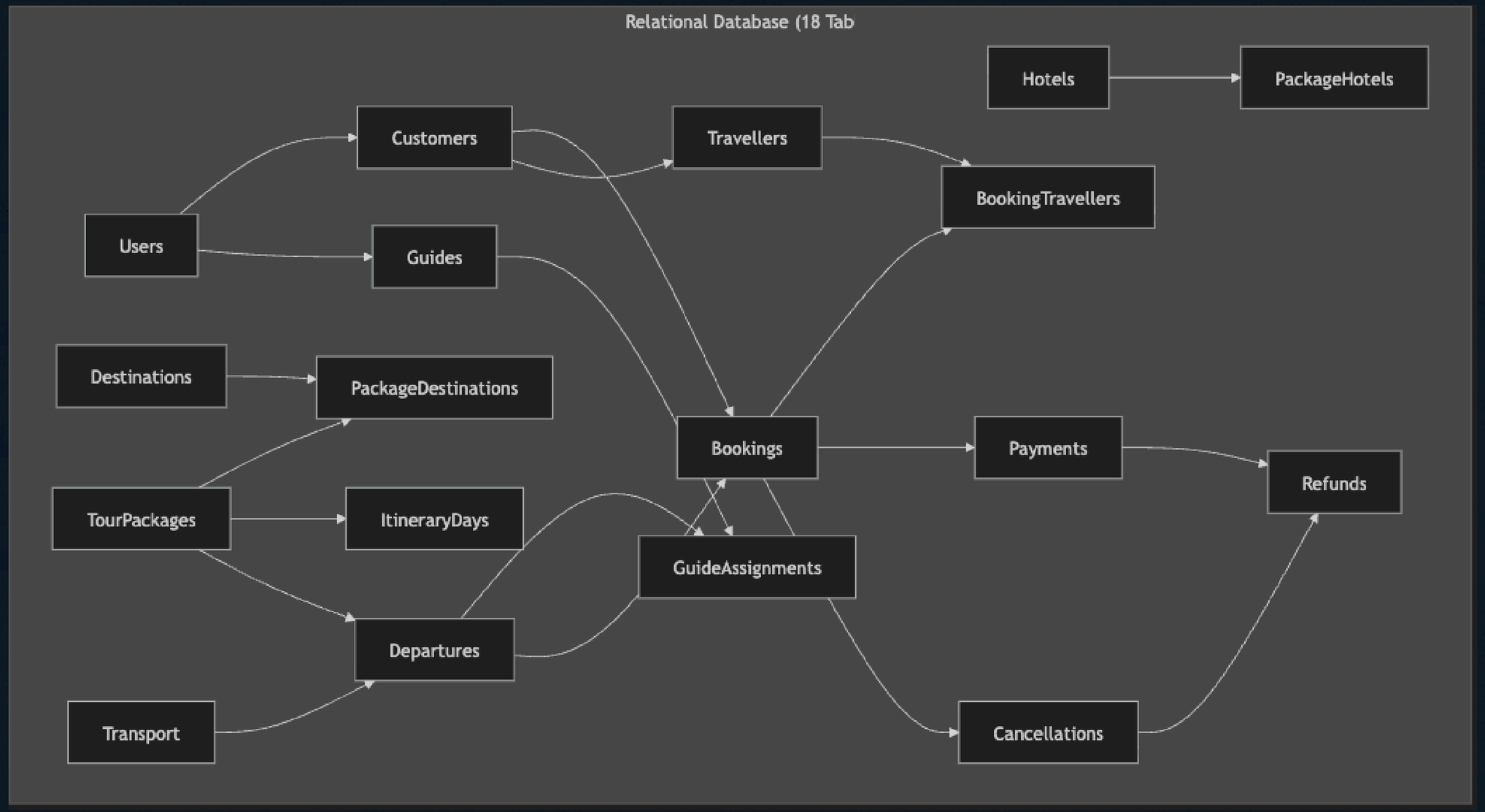
User role	Main responsibilities
Administrator	Manage users, destinations and system settings
Travel Agency Staff	Create packages, itineraries and departures
Booking Staff	Register customers and manage bookings
Tour Manager	Assign guides and check schedules
Accounts Staff	Record payments, cancellations and refunds
Customer / Traveller	View packages and manage booking details

Functional requirements

- Add, update, view and delete permitted records.
- Search packages by destination, date and price.
- Check departure availability before confirming a booking.
- Assign guides based on availability.
- Process payments, cancellations and eligible refunds.
- Generate booking, revenue and destination-demand reports.

Conceptual Design — ER Diagram

The ER diagram represents the major entities and relationships in the Tourism Package and Travel Booking Management System.



Initial Relational Schema

The proposed relational schema converts the ER model into tables.

Table	Important attributes
Destination	destination_id (PK), name, state, country, description
Package	package_id (PK), destination_id (FK), package_name, duration_days, price
Itinerary_Day	itinerary_id (PK), package_id (FK), day_number, activities
Departure	departure_id (PK), package_id (FK), guide_id (FK), start_date, end_date, capacity, status
Hotel	hotel_id (PK), hotel_name, location, category
Transport	transport_id (PK), transport_type, provider, capacity
Guide	guide_id (PK), guide_name, phone, specialization

Table	Important attributes
Customer	customer_id (PK), name, email, phone
Booking	booking_id (PK), customer_id (FK), departure_id (FK), booking_date, status
Traveller	traveller_id (PK), booking_id (FK), traveller_name, age
Payment	payment_id (PK), booking_id (FK), amount, payment_date, status
Cancellation	cancellation_id (PK), booking_id (FK), cancellation_date, reason
Refund	refund_id (PK), cancellation_id (FK), amount, refund_date, status

Additional relationship tables: Package_Hotel, Package_Transport, and Departure_Transport can represent many-to-many relationships where needed.

PK = Primary Key; FK = Foreign Key.

Primary Keys and Relationships

Primary Keys

A Primary Key (PK) is a column that uniquely identifies each record in a table. It cannot contain duplicate or NULL values.
Examples from your project:

Table	Primary Key
Users	user_id
Customers	customer_id
Guides	guide_id
Destinations	destination_id
TourPackages	package_id
Departures	departure_id
Bookings	booking_id
Travellers	traveller_id
Payments	payment_id
Refunds	refund_id

Relationships

Relationships describe how tables are connected using primary keys and foreign keys.

Relationship	Type
TourPackages → Departures	One-to-Many
TourPackages → ItineraryDays	One-to-Many
Customers → Bookings	One-to-Many
Bookings → Payments	One-to-Many
Bookings → Travellers	Many-to-Many, through BookingTravellers
Destinations → TourPackages	Many-to-Many, through PackageDestinations
TourPackages → Hotels	Many-to-Many, through PackageHotels
Payments → Refunds	One-to-Many, if partial refunds are supported